

Declare and initialize a String variable

A **String** in Java is a sequence of characters, and it is a reference data type. Strings are objects of the `String` class, and they are widely used for handling text-based data.

1. Declaration and Initialization

Syntax

```
String variableName = "value";
```

Example

```
String message = "Hello, World!";
```

Here:

- `String` specifies the data type.
 - `message` is the name of the variable.
 - `"Hello, World!"` is the value assigned to the variable.
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2. String Variable Examples

Example 1: Simple Declaration and Initialization

```
public class StringExample {  
    public static void main(String[] args) {  
        String greeting = "Welcome to Java Programming!";  
        System.out.println(greeting);  
    }  
}
```

Output:

```
Welcome to Java Programming!
```

Example 2: Declaring Multiple String Variables

You can declare and initialize multiple String variables on the same line.

```
public class MultiStringExample {  
    public static void main(String[] args) {  
        String firstName = "John", lastName = "Doe";  
        System.out.println("Full Name: " + firstName + " " + lastName);  
    }  
}
```

Output:

```
Full Name: John Doe
```

3. Declaring and Initializing a String with `new` Keyword

You can also create a String object using the `new` keyword.

```
public class StringWithNew {  
    public static void main(String[] args) {  
        String str = new String("Learning Java!");  
        System.out.println(str);  
    }  
}
```

Output:

```
Learning Java!
```

4. Mutability of Strings

Strings are immutable in Java, meaning their values cannot be changed after they are created. Any operation that modifies a String creates a new String object.

5. Best Practices

1. Use meaningful names for String variables to enhance code readability.
2. Prefer string literals (`String str = "value";`) over creating new objects (`String str = new String("value");`) for better memory management.
3. Use `StringBuilder` or `StringBuffer` for scenarios where frequent modifications are needed, as they are mutable.